

WEEK – 13

DURATION: 06 April 2026 – 10 April 2026

DAY 1: 06 April 2026

1. TASKS WORKED ON

- **MB-USA-Trucking Project:** Worked on the setup and workflow analysis of the mb-usa-trucking project while exploring the migration from Cloudinary to AWS S3 for cloud storage. Also investigated and worked on resolving CloudFront verification issues by analyzing solutions and raising a support case.
- **Advanced Form Management:** Learned advanced form handling and validation techniques in React, including file uploads, image previews, and dynamic form field management. Also gained practical experience in handling structured form data and implementing secure input validation.

2. LEARNING AND KNOWLEGDE GAINED

Today, I gained practical knowledge of cloud service migration from Cloudinary to AWS S3 and explored cloud integration challenges. I also strengthened my React skills by implementing form validation, file uploads, and dynamic form field management.

3. PLAN FOR NEXT WORKING DAY

- Learn additional advanced React concepts.
- Continue working on the “mb-usa-trucking” project and cloud integration tasks.

DAY 2: 07 April 2026

1. TASKS WORKED ON

- **Advanced API Integration:** Learned advanced frontend concepts including Axios Interceptors for API handling, TanStack Query for optimistic UI updates, and WebSocket communication for real-time data updates. Implemented practical examples to understand API management, state synchronization, and real-time application behavior.
- **MB-USA-Trucking Project:** Worked on testing the mb-usa-trucking website in the production environment and analyzed the document storage workflow. Also configured the MongoDB production connection and set up Super Admin access with the required permissions.

2. LEARNING AND KNOWLEDGE GAINED

Today, I strengthened my understanding of advanced API integration, real-time communication, and modern state management techniques. I also gained practical experience with production-level backend configurations, document management, and role-based access control.

3. PLAN FOR NEXT WORKING DAY

- Continue learning advanced React concepts.
- Work on additional features and testing in the “mb-usa-trucking” project.

DAY 3: 08 April 2026

1. TASKS WORKED ON

- ***MB-USA-Trucking Website Enhancements:*** Worked on improving the application's UI by refining the layout, profile page, and user dropdown for a better user experience. Also performed production testing to identify and document issues related to functionality, responsiveness, and overall usability.
- **React Optimization Concepts:** Learned about React rendering behavior and the importance of performance optimization in modern applications. Implemented `React.memo` and `useMemo` to reduce unnecessary re-renders and improve application efficiency.

2. LEARNING AND KNOWLEDGE GAINED

Today, I gained practical experience in testing a live production website and identifying issues related to UI, functionality, and performance. I also strengthened my understanding of React optimization techniques using `React.memo` and `useMemo` to improve application efficiency.

3. PLAN FOR NEXT WORKING DAY

- Learn additional React optimization concepts.
- Continue working on the “mb-usa-trucking” website and testing-related improvements.

DAY 4: 09 April 2026

1. TASKS WORKED ON

- ***MB-USA-Trucking – Documentation & Form Validation:*** Worked on documenting form validation logic across different modules to ensure proper data integrity and consistency throughout the project.
- **React Performance Optimization:** Learned advanced React optimization techniques using `React.memo`, `useMemo`, and `useCallback` to improve performance and reduce unnecessary re-renders. Also explored code splitting and lazy loading with `React.lazy` and `Suspense` to optimize application loading speed.
- **Advanced Styling & Animation:** Learned dynamic styling techniques using CSS-in-JS and advanced Tailwind CSS patterns for responsive design and dark mode implementation. Also explored Framer Motion to create smooth and interactive animations for modern user interfaces.

2. LEARNING AND KNOWLEDGE GAINED

Today, I strengthened my understanding of React performance optimization and learned when to apply optimization techniques effectively. I also gained practical experience in building responsive, animated user interfaces and understood the importance of proper validation documentation for maintaining system reliability.

3. PLAN FOR NEXT WORKING DAY

- Continue learning advanced React concepts and optimization techniques.
- Explore additional frontend performance and UI enhancement strategies.

DAY 5: 10 April 2026

1. TASKS WORKED ON

- ***Advanced State Management:*** Learned modern state management techniques by implementing a Todo application using useReducer and managing global application state with Zustand. This helped strengthen my understanding of centralized and scalable state management in React applications.
- ***Error Boundaries & React Concepts:*** Learned advanced React concepts including Error Boundaries, React Portals, Fragments, and Strict Mode. Implemented practical examples to improve error handling, DOM structure management, and application reliability.

2. LEARNING AND KNOWLEDGE GAINED

Today, I strengthened my understanding of functional testing, advanced state management, and React best practices. I also explored concepts such as Zustand, reducers, Error Boundaries, Portals, Fragments, and Strict Mode through practical implementation.

3. PLAN FOR NEXT WORKING DAY

- Continue learning advanced React concepts and testing methodologies.
- Explore additional optimization and state management techniques.